mediatools 1.2.12 – Handbook

Architecture, operation and safety design of shrink, progress, triage, flatten, sync and snapshot

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Abstract

mediatools is a Python package (standard library only) with subcommands for maintaining large local media collections on Linux: `shrink` re-encodes videos to HEVC in place and records them in a per-directory SQLite database; `progress` reports the shrink status read-only; `triage` presents short clips in batches, one player window each, and removes the clips whose windows the user closed together with their hard links; `flatten` creates, per unit, a flat directory of hard links whose names encode the original location within the file-system name limit; `sync` copies a tree to a backup disk with rsync while preserving hard links, never deletes, and reports files that exist only on the target; `snapshot` keeps versioned rsnapshot backups on a removable disk, adopts an existing copy without copying data and refuses implausible runs after a dry-run probe. All machine-specific paths and names live in one local configuration file outside the repository. This handbook describes installation, configuration, the architecture (one class per module), the decision logic and safety rules of every subcommand, the backup strategy, testing and troubleshooting. All behaviour described is covered by the automated tests of the release; the snapshot workflow was additionally run end to end on an ext4 file system.

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1 Overview

1.1 Purpose

mediatools maintains a local copy of videos and pictures that a download tool fetches folder by folder, for example an archiver for a creator platform. Such collections grow into deeply nested trees with large H.264 videos, many unwanted short clips and backups that silently turn hard links into full copies.

The central design rule follows from how download tools work: they decide by **file name** whether a file is already present. mediatools therefore never takes a name away – a shrunk video keeps its name, a discarded clip leaves an empty placeholder, a flat view consists of additional names (hard links), and `sync` never deletes. The next download run fetches only what is really new.

`shrink` re-encodes lossy and replaces the original; the only undo is a backup. The recommended order for a new user is therefore: configure, look (`progress`, `--dry-run`), judge the quality with `shrink --keep-copies` on a test folder, back up, then work (section “First steps”).

1.2 Subcommands

Subcommand	Purpose	Writes to
`shrink`	HEVC re-encode in place, name kept	original files (same inode), `shrink.db`
`progress`	which videos are recorded as done	data directory only
`triage`	review short clips, discard by closing the window	trash (or delete), kept list
`flatten`	flat hard-link view per unit	`<unit>/_flat/` only
`sync`	backup preserving hard links, no deletions	target directory, reports
`snapshot`	versioned backups with rsnapshot	snapshot root on the backup disk
`config`	show or create the configuration file	data directory only
`rename-db`	rename shrink databases to the configured name	database files only

Shared rules, implemented once in `mediatools.core`:

- `_flat`, `newShrinkedFiles` and hidden directories are never entered by a scan (`TreeWalker`, `MediaTypes.SKIP_DIRS`).
- Symbolic links are never followed or processed.
- Every run writes `logs/<command>_<timestamp>.log` into the data directory `$MEDIATOOLS_HOME` (default `~/scripts/mediatools.d`).
- External programs are called with argument lists, never through a shell, so file names with spaces, quotes or `$` are safe.

2 Installation

Everything except the system packages runs as a **normal user**.

2.1 Prerequisites

```
sudo apt install ffmpeg vlc libglib2.0-bin rsync rsnapshot git python3-venv
ffmpeg -hide_banner -encoders | grep libx265
```

`sudo` is needed because apt writes to `/usr` and the package database; from the user context it is sufficient, a root shell is not required. Only the packages of the subcommands in use are needed (README, table “System packages”). The second command runs as normal user and must print a line containing `libx265`.

2.2 Option A: installer (recommended)

mediatools is installed from a **release tag** (`vX.Y.Z`) on GitHub, never from the working files of a clone, so `mediatools --version` always shows the installed release:

```
curl -fsSLO https://raw.githubusercontent.com/peddanet/mediatools/main/install.sh
bash install.sh
mediatools --version
```

- `curl -fsSLO URL` – download the installer into the current directory (`-f` fail on HTTP errors, `-sS` quiet but show errors, `-L` follow redirects, `-O` keep the remote file name). A clone contains the same script; running it from there makes no difference, because it installs from GitHub, not from the clone.
- `bash install.sh` – reads the release tags with `git ls-remote`, picks the newest `vX.Y.Z` (or the tag given as argument), installs it with pip into `~/scripts/mediatools.d/venv`, checks that the program reports the tag's version and links `~/scripts/mediatools` to it. It replaces only a link it created itself or the launcher of version 1.1.0 and refuses any other file.
- `bash install.sh --list` – list the releases; `bash install.sh v1.2.0` – install a given release, also an older one.
- `BIN_DIR=~/.local/bin bash install.sh` – link the command elsewhere.

Everything runs as normal user; the installer writes only below `$MEDIATOOLS_HOME` and `BIN_DIR`. It needs `git` and the venv module (`sudo apt install git python3-venv`).

Update: run `bash install.sh` again. Remove: `rm -rf ~/scripts/mediatools.d/venv ~/scripts/mediatools`.

2.3 Option B: pipx

```
pipx ensurepath
pipx install git+https://github.com/peddanet/mediatools.git@v1.2.0
```

- `pipx ensurepath` – adds `~/.local/bin` to `PATH` (once; open a new terminal afterwards).
- `pipx install git+URL@TAG` – builds the given release into a virtual environment of its own; replace `v1.2.0` with the newest tag. Update with `pipx install --force git+URL@NEWTAG`, remove with `pipx uninstall mediatools`.

Ubuntu 24.04 and later block `pip install --user` for the system Python (PEP 668); both options use a virtual environment instead.

2.4 Development checkout

Code imported from a clone (development venv with `pip install -e`) is marked by `--version`:

```
mediatools 1.2.0 (source checkout /home/USER/Projects/mediatools, not an installed release)
```

The installed release is not affected by anything in the clone.

2.5 Data directory

Configuration, logs, state, reports and the installed release (`venv/`) are kept in `$MEDIATOOLS_HOME`, default `~/scripts/mediatools.d`. Another location is set in `~/.profile` before installing:

```
echo 'export MEDIATOOLS_HOME="$HOME/.local/share/mediatools"' >> ~/.profile
```

2.6 First steps

Nothing in the first block changes a file:

```
mediatools config --set collection.root=~/Videos/collection
mediatools config
mediatools progress
mediatools triage --dry-run --max-seconds 30
mediatools flatten --dry-run
```

Judge the shrink quality on a copy (`--keep-copies` leaves the originals untouched and records nothing; a smaller `--crf` means higher quality):

```
mkdir -p ~/shrinktest && cp ~/Videos/collection/projectA/some/video.mp4 ~/shrinktest/
mediatools shrink --keep-copies ~/shrinktest
```

Then back up and work; every step asks before it changes anything:

```
mediatools sync
mediatools triage --max-seconds 30
mediatools shrink
mediatools flatten
```

2.7 Running the tests on the target machine

```
cd ~/Projects/mediatools
sudo apt install python3-pytest
python3 -m pytest -q
```

Expected: all tests pass. Tests that need ffmpeg, rsync or rsnapshot are skipped when the tool is missing. `sudo` is needed only for the apt installation; the tests run as normal user.

3 Configuration

Everything that describes a concrete collection – paths, the database name, folder names worth abbreviating, the backup disk – is kept in one INI file outside the repository, so neither the code nor its history allows conclusions about the data:

```
$MEDIATOOLS_HOME/mediatools.ini      default: ~/scripts/mediatools.d/mediatools.ini
```

```
mediatools config --init        # write the commented template (never overwrites)
mediatools config --path        # print the file location
mediatools config               # print the effective values incl. defaults
```

Section / key	Default	Used by
`[collection] root`	–	every command called without a path
`[sync] source`	`[collection] root`	`sync` called without a source
`[shrink] db_name`	`shrink.db`	shrink, progress, rename-db
`[flatten] abbreviations`	`Videos=V, Images=I, Audios=U`	flatten, naming level 1
`[snapshot] source, disk`	–	snapshot (required)
`[snapshot] root`	`rsnapshot`	directory on the disk
`[snapshot] retain_*`	alpha 7, beta 4, gamma 12, delta 2	snapshots per level; 0 disables a level
`[snapshot] exclude`	`newShrinkedFiles/`	rsync exclude patterns
`[snapshot] max_deletions`	500	abort threshold of a run

A missing or invalid value stops the command with a message that names the key and the expected format.

Which directory is used. A path on the command line always wins; the file only provides defaults. Different jobs usually start at different levels – for example the collection for `shrink`, `triage` and

`flatten`, but the whole home directory for `sync` and `snapshot` – hence the separate keys. Change a default without an editor:

```
mediatools config --set collection.root=~/Pictures/collection
mediatools config --set sync.source=~ --set snapshot.source=~
mediatools config                      # show the effective values
```

`--set SECTION.KEY=VALUE` replaces only that line and keeps all comments; a missing file is created from the template first.

For `flatten`, `root` must be the **master level**: each direct sub-directory is treated as one unit with its own `_flat`.

3.1 Renaming existing databases

If the shrink databases of an existing collection carry another file name, set `[shrink] db_name` and rename them in one pass. The old name is passed on the command line only:

```
mediatools rename-db OLDNAME.db NEWNAME.db PATH --dry-run     # list, change nothing
mediatools rename-db OLDNAME.db NEWNAME.db PATH               # rename
```

`OLDNAME.db` and its backups `OLDNAME.db.bak_*` become `NEWNAME.db` and `NEWNAME.db.bak_*`; there is no further question. Afterwards `NEWNAME.db` is stored as `[shrink] db_name` (comments in the file are kept), so `shrink` and `progress` find the renamed databases. `PATH` defaults to `[collection] root` and may also follow the options. Renaming within a directory keeps the inode; content and time stamps are unchanged. A directory that already contains the new name is reported as a conflict and left alone – nothing is overwritten.

4 Architecture

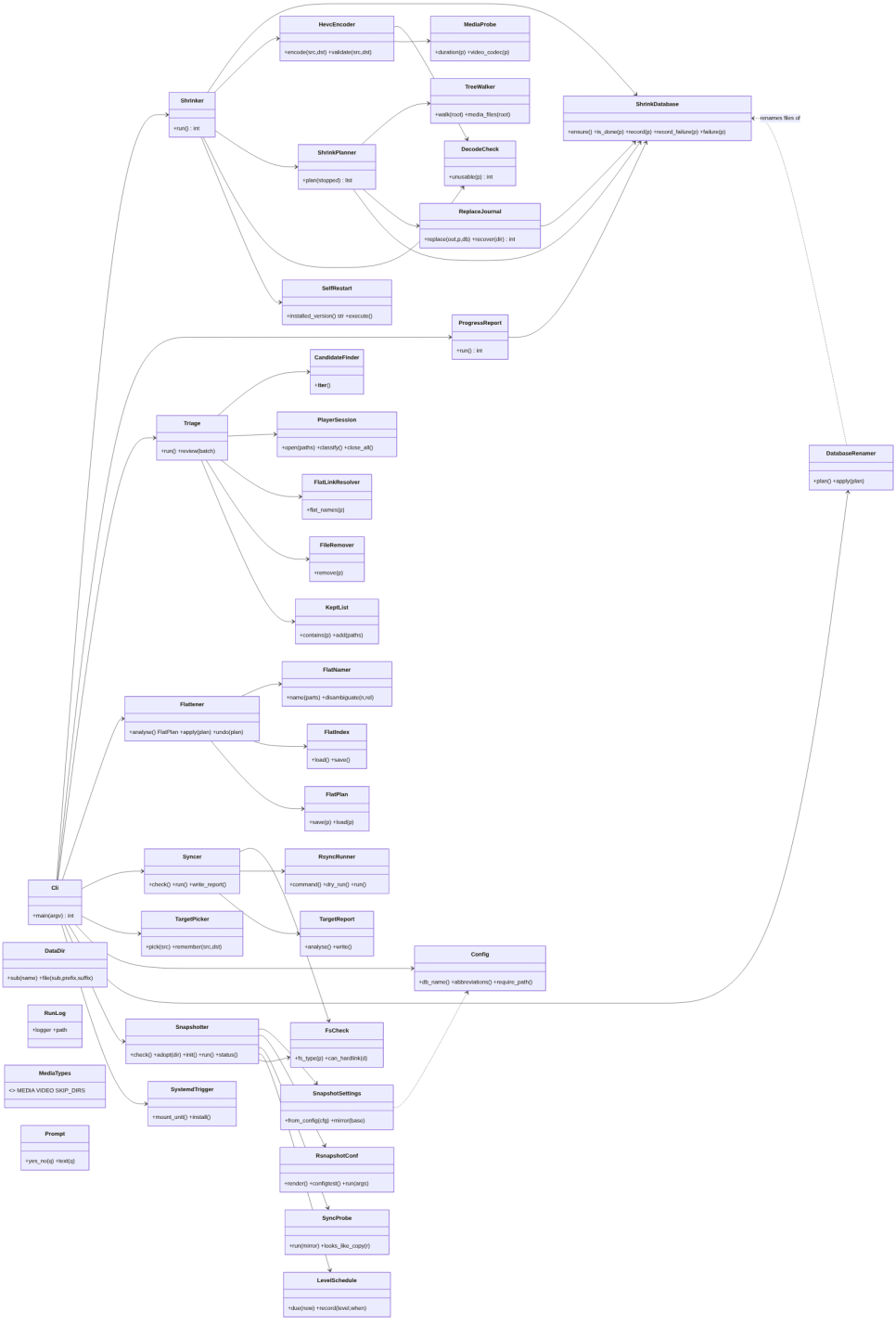


Figure 1: Classes and dependencies (one class per module)

```
src/mediatools/
  cli.py               Cli
  core/
    config.py          Config           local INI file (paths, names)
    datadir.py         DataDir          location of logs, state, plans, reports
    runlog.py          RunLog           logger to terminal and time-stamped file
    runcontrol.py      RunControl       PID/stop/status files, background mode
    runstatus.py       RunStatus        mediatools status: running release, advice, log
    attachview.py      AttachView       live view of a background run (attach)
    sleepinhibitor.py  SleepInhibitor   systemd sleep/idle lock during a run
    progressbar.py     ProgressBar      two progress lines, status file
    mediatypes.py      MediaTypes       extensions, reserved directory names
    probe.py           MediaProbe       ffprobe wrapper (duration, codec)
    decodecheck.py     DecodeCheck      trial decode: finds videos that cannot be decoded
    selfrestart.py     SelfRestart      switch a running job to the installed release
    walker.py          TreeWalker       sorted os.walk with skip rules
    prompt.py          Prompt           yes/no and text questions, --yes support
  shrink/
    database.py        ShrinkDatabase   per-directory database, schema migration
    dbrenamer.py       DatabaseRenamer  one-time renaming of databases
    encoder.py         HevcEncoder      ffmpeg call and result validation
    shrinker.py        Shrinker         per-file decision, failures, summary
    planner.py         ShrinkPlanner    planning: DB, failures, durations, codecs, order
    journal.py         ReplaceJournal   crash-safe copy-back with marker file
    shrinkstate.py     ShrinkState      caches, throughput, result table of a run
    fileresult.py      FileResult       outcome of one video
    resulttable.py     ResultTable      framed result table, report file
    progress.py        ProgressReport   read-only done/todo lists
  triage/
    candidates.py      CandidateFinder  lazy search by duration
    player.py          PlayerSession    one process per clip, classification
    linkresolver.py    FlatLinkResolver _flat names of a file (by inode)
    remover.py         FileRemover      gio trash or permanent delete
    keptlist.py        KeptList         decisions across sessions
    triage.py          Triage           batch workflow
  update/
    updater.py         ReleaseUpdater   mediatools update (runs apply-release.sh)
    apply-release.sh   -                applies delivered patch archives
  flatten/
    namer.py           FlatNamer        names within the byte limit
    flatindex.py       FlatIndex        _flat/_index.tsv
    planentry.py       PlanEntry        one planned action
    plan.py            FlatPlan         actions, TSV persistence, undo input
    flattener.py       Flattener        analyse / apply / undo
  sync/
    fscheck.py         FsCheck          fs type, hard-link test, nesting
    target_picker.py   TargetPicker     interactive disk choice, remembered
    rsync_runner.py    RsyncRunner      rsync command, dry-run parsing
    target_report.py   TargetReport     target-only files, duplicate check
    syncer.py          Syncer           checks, dry run, transfer, report
  snapshot/
    settings.py        SnapshotSettings validated [snapshot] values
    rsnapconf.py       RsnapshotConf    generated rsnapshot.conf, configtest
    syncprobe.py       SyncProbe        rsync dry run: volume, removals
    schedule.py        LevelSchedule    which levels are due
    snapshotter.py     Snapshotter      check / adopt / init / run / status
    trigger.py         SystemdTrigger   user unit started by the disk mount
```

Design principles:

- **One responsibility per class**, every public class and method has a docstring (shown with `python3 -m pydoc mediatools.flatten.namer`).
- **Dependency injection** – logger, prompt, probe, player and remover are constructor arguments; the tests replace them with fakes.
- **Plan before action** where actions are irreversible or numerous (`flatten`, `sync`, `triage`): analysis, summary, confirmation, execution.

5 shrink

5.1 Invocation

```
mediatools shrink ~/Pictures/collection                        # largest first
mediatools shrink ~/Pictures/collection --min-seconds 60       # skip short clips
mediatools shrink ~/Pictures/collection --order path --keep-copies
mediatools shrink ~/Pictures/collection --crf 26 --threads 8 -y
mediatools shrink ~/Pictures/collection --foreground           # stay in this terminal
mediatools attach                                              # (alias: front) watch the backgro
mediatools stop                                                # finish the current file, then en
mediatools reload                                              # switch a running shrink to the i
mediatools status                                              # running? release, current or not
```

Option	Default	Meaning
`--order`	`largest`	`largest`, `smallest` or `path` (directory order)
`--min-seconds S`	0	leave out videos shorter than S seconds
`--keep-copies`	off	do not touch originals; keep copies in `newShrinkedFiles/`; record nothing
`--retry-failed`	off	include videos that failed in an earlier run (`FAILED`/`BROKEN`)
`--crf N`	28	libx265 constant rate factor (lower = better quality, larger)
`--threads N`	CPUs - 1	decoder threads and size of the x265 thread pool
`--ext LIST`	`mp4`	extensions to process
`-y`	off	no confirmation question
`-v`	off	debug logging

5.2 Plan

Before anything is encoded, the whole tree is walked once:

1. Interrupted replacements are completed (see *Interruption*); remaining `newShrinkedFiles/` folders are noted as leftovers of aborted runs.
2. Every video is checked against the database of its directory; recorded files are counted as processed and left out. 0-byte placeholders are ignored. **Exception – regrown:** a recorded video that is now at least 1.2 times as large as every size recorded for its name, and whose video stream is not HEVC, is planned again. This is what happens when an original is copied back over its shrunk version, e.g. by a sync from the backup disk in the wrong direction; the plan summary reports the number (`recorded as shrunk, but larger again and not HEVC`). A video recorded as failed in an earlier run (`FAILED` or `BROKEN`, table `failures`) is left out

while its size is unchanged; the plan summary and the end of the run list it as `FAILED BEFORE`. `--retry-failed` includes these videos again; a successful retry deletes the failure row.

3. With `--min-seconds`, the durations of the remaining videos are read (cached in `state/triage_durations.ts` shared with `triage`) and short ones are left out.
4. The codec of every remaining video is read (cached in `state/shrink_codecs.tsv`); videos that are already HEVC are set aside. They are recorded after the start question without encoding and are not counted in volume, time estimate and progress.

   Both caches are keyed by path, size and modification time: a value is reused as long as the file is unchanged, so only new or changed videos are probed again. New values are written every 500 files, so an interrupted first check keeps its work. The messages show the share answered from the cache, e.g. `checking codecs: 1000 of 3076 (998 from cache)`.
5. The rest is sorted (`--order`), and the run reports, for example (numbers illustrative):

```
already processed (database): 12000 | shorter than 60 s: 1500 | already HEVC,
recorded without encoding: 40 files, 85.0 GiB
to encode: 6500 files, 560.0 GiB, about 4 d 15 h at the measured 1.5 MB/s
expected afterwards: about 215.6 GiB, saving about 344.4 GiB (output 38 % of the
source, measured on 25 file(s))
leftover work folders of aborted runs: 18 folder(s), 3 file(s), 0.4 GiB - deleted
after confirmation
Start encoding and REPLACE the originals? [y/N]
```

The estimates use `state/shrink_throughput.json`: source bytes and encoding seconds (speed) and, since 1.2.1, source and output bytes of the encoded files (size ratio). Before the first measured file the line says “not yet known”. During the run the measured speed of the run itself takes over. Leftover folders are listed one by one with `-v`; answering “no” keeps them.

5.3 Decision per file

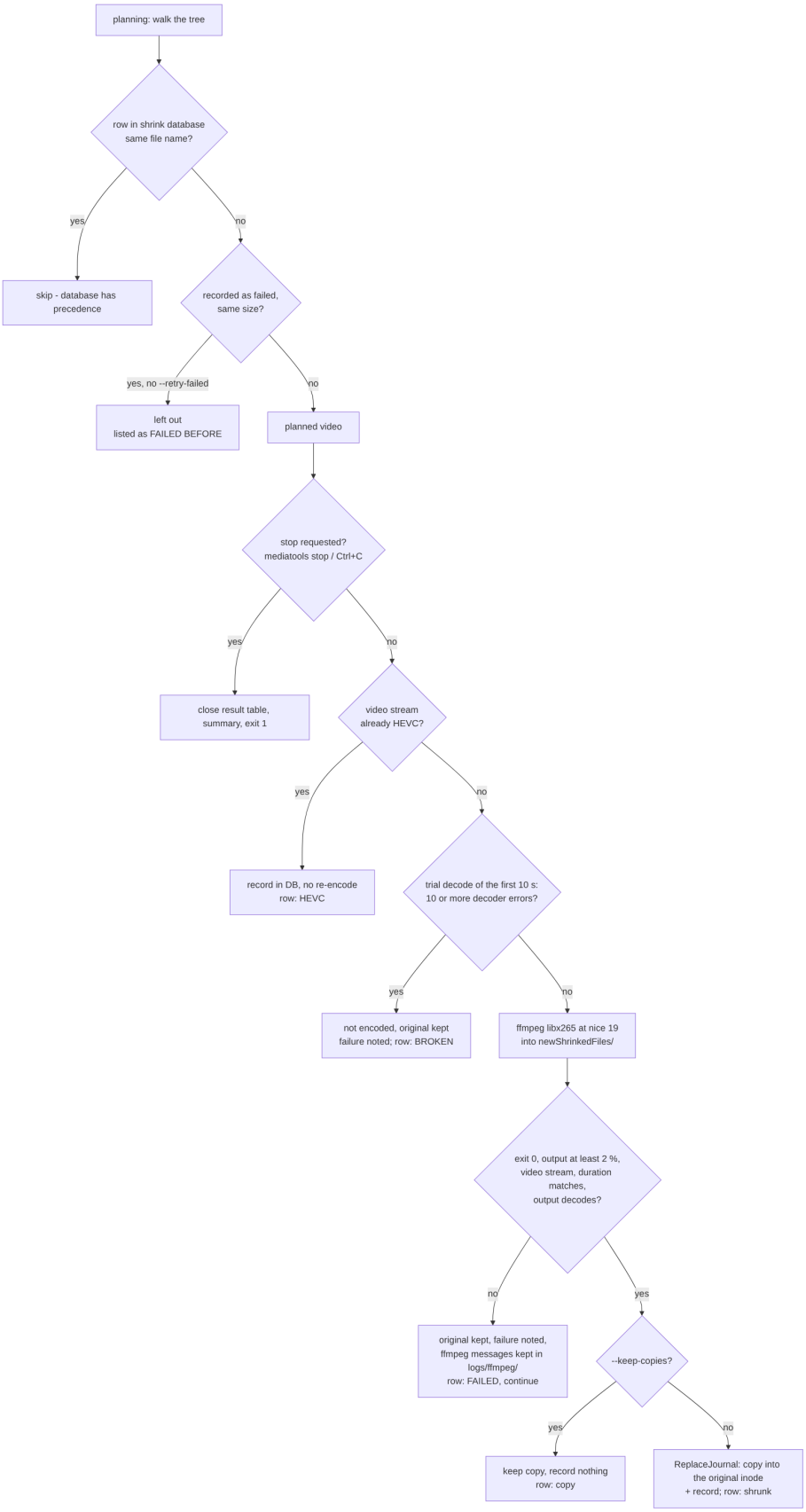


Figure 2: Decision logic of Shrinker

1. **Database first.** A recorded file name means “already processed”, regardless of the codec it was processed with; the file is skipped without probing.
2. **Codec.** If the first video stream is already HEVC, the file is recorded but not re-encoded (re-encoding HEVC costs quality for little gain).
3. **Encode and validate.** `HevcEncoder` runs

   ```
   nice -n 19 ffmpeg -nostdin -y -hide_banner -nostats -progress pipe:1 \
          -threads N -i SRC -vcodec libx265 -crf 28 -x265-params pools=N TMP
   ```

   `N` is the number of logical CPUs minus one (`--threads`). `-threads` before `-i` sets the decoder threads; libx265 runs its own thread pool, which ffmpeg’s `-threads` does not limit and which would otherwise use every CPU, so `pools=N` caps it. One CPU stays free for the desktop. `nice -n 19` gives ffmpeg the lowest CPU priority: it still uses idle CPUs, but yields at once to interactive programs.

   in its own process group and accepts the result only if the exit code is 0, the output is not empty and at least 2 % of the input’s size, has a video stream, its duration equals the input within `max(1 s, 2 %)`, and **the whole output decodes without a single error line** (`ffmpeg -v error -i OUT -f null -` at `nice` 19; time limit `max(600 s, 2 × duration)`; a timeout or a missing ffmpeg counts as failure). Measured cost: a full decode takes about 5–10 % of the encode time of the same file. A tiny or undecodable output is typical of an input whose data the decoder could not use – the encoder then writes little more than grey frames. ffmpeg’s messages go to `TMP.ffmpeg.log`; on failure its last lines are logged and the complete file is kept as `logs/ffmpeg/<timestamp>_<name>.ffmpeg.log` in the data directory.
4. **Copy back in place.** `shutil.copyfile` writes into the existing inode. All hard links of the file – `_flat` entries, deduplication links – therefore show the shrunk content immediately, and no second encode happens via another name.

5.4 Progress display

(numbers illustrative)

```
total [##############..........]  37.5 % | 2500/6500 files | 210.0/560.0 GiB | ETA 2d21h
file   63 % of 1.20 GiB (0.76 GiB encoded) | 8m left | clip.mp4
```

The first line covers the whole run, the second the file being encoded: its percentage, its size, the part already encoded and its remaining time at the current speed. Both lines are redrawn in place and cut to the terminal width (a long file name is shortened in the middle), so a narrow window does not fill up with lines.

Percentage, volume and remaining time are measured in source bytes, because the encoding time grows with the file size, not with the number of files. Files that need no encoding – HEVC by now, e.g. a second hard link of a file shrunk earlier in the run – are removed from the totals instead of being counted as done, so they do not inflate progress and speed. The lines are drawn only on a terminal; the log file receives the normal messages.

5.5 Result table

Each finished video adds one row to a framed table: row number, time of completion, size before and after, ratio, playing time, wall-clock time spent, status (`shrunk`, `copy`, `HEVC`, `FAILED`, `BROKEN`) and the path relative to the shrink root. The figures have fixed columns; the path takes the rest of the width and wraps inside its cell, after a `/` or a space, so a long name makes the row taller instead of breaking the frame:

```
results (also in reports/shrink_results_<ts>.txt):
```

no	end	source	result	ratio	length	took	status	file
1	10:25	2815 MB	284 MB	10.1 %	26.5 min	21.3 min	shrunk	unit_a/ Videos/ long clip name.mp4
2	10:25	2303 MB	-	-	60.0 min	0.5 min	FAILED	unit_b/ clip.mp4

Copy	Width	Where
terminal (and log file)	terminal width, at least 90 columns	rows printed as files finish
report	160 columns	`reports/shrink_results_<ts>.txt`, written and flushed per row

The report is complete up to the last finished file, even after a crash. The details behind a row (ffmpeg command, validation, database record) are written to the log file only; `--verbose` also shows them on the terminal. Errors (e.g. the last lines of a failed ffmpeg run) stay visible.

5.6 Sleep and idle lock

`shrink` takes a systemd inhibitor lock for its whole run, the same as wrapping it in `systemd-inhibit --what=sleep:idle`:

```
systemd-inhibit --list     # lists: mediatools ... mediatools shrink  sleep:idle  block
```

The lock is held by a helper process (`systemd-inhibit ... tail --pid=<shrink pid> -f /dev/null`) in its own session: Ctrl+C in the terminal does not reach it, and it ends at the latest when the shrink process ends, even after `kill -9`, so no lock is left behind. No root rights are needed in a local session: the polkit defaults of systemd allow `org.freedesktop.login1.inhibit-block-sleep` for active and inactive local sessions and `-idle` for everyone (see `/usr/share/polkit-1/actions/org.freedesktop.login1.policy`). If the lock is refused or `systemd-inhibit` is missing, shrink logs a warning and runs anyway.

5.7 Database

```
CREATE TABLE files (path text NOT NULL PRIMARY KEY,
                    size integer, modified text);
CREATE TABLE failures (name text NOT NULL PRIMARY KEY, size integer,
                       status text, reason text, at text);
```

- One database per directory with videos; file name `[shrink] db_name` (default `shrink.db`).
- **Matching uses the file name**, not the stored absolute path: a database describes the files of its own directory, and the stored path still points to wherever the collection was when the row was written. A row whose size differs from the file today is reported by `progress` as “size changed since” and still counts as processed – nothing is encoded twice. Only a file that is at least 1.2 times as large as recorded (**regrown**) and not HEVC is encoded again.
- `path` and `modified` (mtime after shrinking, `YYYY-MM-DD HH:MM:SS`) are still written for compatibility and traceability.

- `failures` holds videos that could not be shrunk (`FAILED`, `BROKEN`) with their size, status, cause and time. It is kept apart from `files`, so `files` still means "processed" for every tool that reads it; older versions ignore the extra table. A row applies only while the file has the recorded size. Review mode (`--keep-copies`) writes no rows.
- Databases of the first script generation (no `PRIMARY KEY`) are backed up next to the original (suffix `.bak_` plus timestamp) and migrated in one transaction.

Inspection (normal user, read-only):

```
sqlite3 -readonly -header -column ~/Pictures/collection/A/Videos/shrink.db \
  "SELECT size, modified, path FROM files ORDER BY modified;"
sqlite3 -readonly -header -column ~/Pictures/collection/A/Videos/shrink.db \
  "SELECT at, status, reason, size, name FROM failures ORDER BY at;"
```

5.8 Interruption and failures

Event	Behaviour
`mediatools stop` in any terminal	the current file is finished, validated, copied and recorded; then the run stops
terminal closed (background mode, the default)	nothing - the run has its own session and continues
Ctrl+C (or Ctrl+Z) once, `--foreground`	the same as `mediatools stop`
Ctrl+C a second time, SIGTERM, SIGHUP (terminal of a `--foreground` run closed)	ffmpeg is stopped at once; its partial output is deleted; the original stays untouched
power failure, kill -9	see the replacement journal below
ffmpeg cannot encode a file, or the output fails validation	logged as `FAILED`, failure row written, ffmpeg messages kept, run continues, exit code 2
video data cannot be decoded (still encrypted, scrambled)	set aside as `BROKEN` before encoding: not encoded, original untouched, failure row written; exit code unchanged
a video failed in an earlier run, size unchanged	left out, listed as `FAILED BEFORE`; `--retry-failed` includes it
`mediatools shrink` while another shrink runs	refused at once with the PID and the hints `attach` / `stop`, exit code 3

Background mode. Planning and the confirmation happen in the terminal. Then (`RunControl.begin_work`) the process forks: the parent prints how to follow and stop the run and exits, the prompt returns. The child calls `setsid()` - a new session without controlling terminal, so closing the window sends it no SIGHUP - reads from `/dev/null`, writes its terminal output (normally empty; e.g. a Python traceback after a crash) to `state/shrink.console` and registers its PID in `state/shrink.pid` together with the log file path. Only then does it take the sleep lock, so the lock belongs to the process that does the work.

```
shrink continues in the background (pid 12345); the terminal may be closed.
  watch: mediatools attach   (Ctrl+C leaves the view)
  stop:  mediatools stop     (after the current file)
  log:   /home/user/scripts/mediatools.d/logs/shrink_2026-11-09_11-30-00.log
```

`mediatools attach` (alias `front`) prints the last 30 log lines, then every new line, and redraws the two progress lines, which the run writes to `state/shrink.status` once per second. Ctrl+C leaves the view; the run continues. The view ends by itself when the run ends. `--foreground` keeps the previous behaviour (run bound to the terminal, Ctrl+C stops).

`time mediatools shrink` measures only the part until the run moves into the background; the log file has the times of the whole run.

Stopping. `mediatools stop` works from any terminal and does not depend on keyboard focus:

```
mediatools stop
# stop requested: shrink (pid 12345) finishes its current file and ends
```

The command checks that a shrink run is alive (PID file `state/shrink.pid` in the data directory), creates `state/shrink.stop` and returns at once. The run checks for this file every 2 seconds, shows `STOPPING after this file` at the start of its first progress line, completes the current file and ends with exit code 1. If no run is active, the command prints `no shrink run is active - nothing to stop`, returns 1 and leaves no file behind, so a later run is not stopped by mistake. A stop file left by a crash is removed when the next run starts.

Only one run. While a shrink run is alive, a second `mediatools shrink` is refused before it plans anything:

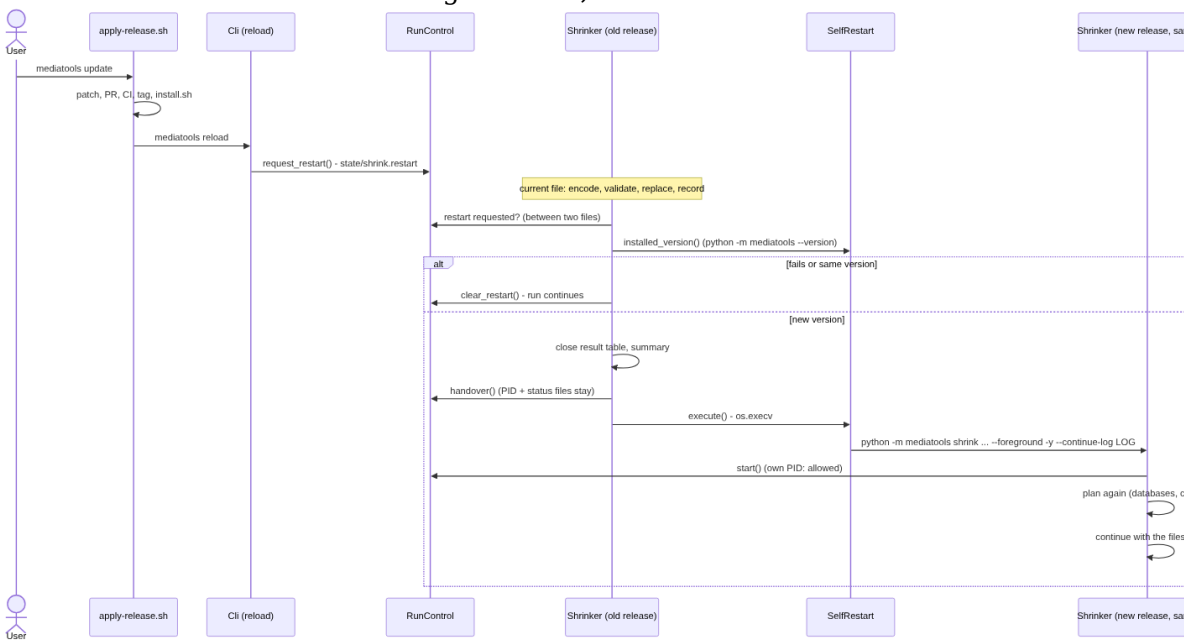
```
shrink is already running (pid 12345) - watch it with 'mediatools attach', stop it with 'mediato
```

A PID file left by a crashed run does not block: only a living process counts.

New release during a run. `mediatools update` installs a new release and then calls `mediatools reload`; the command can also be given by hand after any installation. The running shrink finishes its current file and then switches, without ending the run:

1. Between two files – never during an encode or a replacement – it finds the request file `state/shrink.restart`.
2. It starts the installed release once with `--version`. If that fails, or the version is the one already running, the request is dropped, a line is logged and the run continues unchanged.
3. Otherwise it closes its result table, logs its summary and the line `switching from mediatools A to B now`, and replaces itself (`os.execv`) by `python -m mediatools shrink` of the same installation with the same options plus `--foreground -y`. The process keeps its PID, session and console file; `mediatools attach` and `mediatools stop` keep working, the log file is continued.
4. The new release plans again. Every file finished so far is in the databases, durations and codecs come from the caches, so planning takes seconds and the run continues with the files left. It asks no question (the run was confirmed) and starts a new result table.

The first switch is possible from a running 1.2.10 on: a run of an older release does not know the request and simply finishes with its own code – nothing is mixed, because mediatools loads all its modules at start.



`mediatools reload` without a running shrink prints `no shrink run is active - nothing to reload` and returns 1.

Which release is running? `mediatools status` answers it in one command (normal user, reads only):

```
shrink: running (pid 459729)
running:   1.2.12  (started 2026-11-10 03:21:47)
installed: 1.2.15
status:    'mediatools reload' switches to the installed release after the current file
log:       /home/<user>/scripts/mediatools.d/logs/shrink_2026-11-10_03-21-47.log
pending:   reload requested, takes effect after the current file
```

The running release is the last line `mediatools X (pid N)` in the log of the running job, the one the PID file names; a switch appends such a line to the same log. Older log files keep the release of their own run and say nothing about the current one. The status line says what to do: `up to date`, `'mediatools reload' ...` (running release 1.2.10 or newer), or `X cannot switch releases: 'mediatools stop', ...` (older than 1.2.10). Exit code 0 while a shrink runs, 1 otherwise. `mediatools update` prints this report at its end.

Because ffmpeg runs in its own process group, a Ctrl+C in the terminal reaches only mediatools, which decides what happens. The signal handler only sets flags and never raises an exception, so a replacement in progress always completes.

Replacement journal. Overwriting an original is the only moment in which data could be lost. The sequence is:

1. flush the validated output to disk (`fsync`),
2. write the marker `newShrinkedFiles/<name>.replace` and flush it,
3. flush the directory `newShrinkedFiles/`, so that the marker's directory entry is on disk as well,
4. copy the output into the original (same inode),
5. flush the original,
6. record it in the database,
7. delete the output, delete the marker.

Steps 1-3 guarantee that a marker found after a power failure always comes with a complete output; without them the file system could have written the first part of the copy (step 4) while the marker or the output was still only in memory. If the machine fails anywhere between steps 4 and 7, the next run finds the marker next to the output and repeats the copy before it plans anything else; repeating it is harmless. Without a marker, the original was never touched.

Exit codes: 0 done, 1 stopped by the user, 2 at least one file failed, 3 another shrink run is active.

6 progress

```
mediatools progress ~/Pictures/collection
```

(numbers illustrative)

```
videos: 20000 | processed: 12000 (of which size changed since: 12) |
regrown: 40 | todo: 7960 | to encode: 640.0 GiB
databases: 900 | rows without a matching file: 40
estimated encoding time at the measured 1.5 MB/s: about 5 d 7 h
```

- **processed** – the file name is recorded in the database of its directory; **size changed since** – recorded, but the file has another size today (still counted as processed);
- **regrown** – recorded, but now at least 1.2 times as large (an original restored over its shrunk version); `shrink` encodes it again unless it is already HEVC;

- **todo** – not recorded; **to encode** is the volume of todo and regrown together;
- **rows without a matching file** – the file was deleted or renamed after it was recorded.

The four lists go to `reports/progress_{done,changed,regrown,todo}_<ts>.txt`. The command creates no database and changes nothing.

7 triage

7.1 Workflow

```
mediatools triage ~/Pictures/collection --dry-run
mediatools triage ~/Pictures/collection/A --batch 3
mediatools triage ~/Pictures/collection --min-seconds 10 --max-seconds 30
```

1. **Count first.** `CandidateFinder` collects all clips within the duration range that are not on the kept list; empty placeholders are skipped. Durations come from `state/triage_durations.tsv` when size and modification time are unchanged, otherwise from ffprobe. The session then reports, for example (numbers illustrative):

```
412 of 18230 videos match: 42 batch(es) of up to 10, total clip length 196 min,
estimated review time about 75 min
CLOSE a window = REMOVE that video (an empty placeholder keeps its name);
leave it OPEN = KEEP it.
Start the review? [Y/n]
```

The estimate assumes that each batch is watched once through its longest clip plus 15 s for the decision. `--dry-run` prints the list and this report and stops; `-y` skips only this start question. 2. `PlayerSession` starts one player process per clip (VLC muted and looping by default), then waits 3 s and flags processes that already ended. 3. **Close the window of every video to remove; leave the windows of the videos to keep open.** Then press Enter (`s` skips the batch, `q` quits). 4. Classification and summary, then confirmation – always, also with `-y`:

Process state	Meaning	Action after y
running (window still open)	keep	close window, add to kept list
ended with 0 after the start window (window closed)	remove	remove file and its `_flat` links, leave placeholder
ended within 3 s or with code ≠ 0	player error	keep, not added to kept list

7.2 Placeholders

A removed video is replaced by an empty file (0 bytes) with the same name. Download tools that skip files whose name already exists therefore do not fetch it again. Details:

- The original goes to the trash (or is deleted with `--delete`); the placeholder is created right afterwards and never overwrites a file.
- `shrink`, `progress`, `flatten` and `triage` ignore 0-byte files.
- `sync` passes `--min-size=1` to rsync: the placeholder is not copied over the full file on the backup, and because the placeholder is newer, `-u` prevents a reverse sync from restoring the old file over it.
- `snapshot` treats the placeholder like any change: the newest snapshot holds the empty file, older snapshots keep the video.
- Every removal is appended to `state/triage_discarded.tsv` (`time<TAB>size<TAB>path`).
- `--no-placeholder` removes the name completely.

Whether a particular downloader treats a 0-byte file as “already present” or as an incomplete download is a property of that downloader; test it with a single file before a large session.

7.3 Why `_flat` links are removed too

A hard link keeps the data alive. Removing only the original name of a clip while its `_flat` link still exists frees no space. `FlatLinkResolver` finds, by inode, all names of the file in the `_flat` directories of its ancestors; they are removed together. Hard links outside `_flat` (e.g. from a deduplication tool in another unit) are not touched; the summary shows `[!N further hard link(s) elsewhere: space not freed]`.

7.4 Recovering

```
gio trash --list
gio trash --restore "trash:///x.mp4"
```

Trashed files occupy space until the trash is emptied (`gio trash --empty`).

Option	Default	Meaning
`--max-seconds` / `--min-seconds`	60 / 0	duration range
`--batch`	10	windows per batch
`--player`	`vlc`	e.g. `mpv`, `"flatpak run org.videolan.VLC"`
`--audio`	off	play with sound
`--delete`	off	permanent deletion instead of trash
`--no-placeholder`	off	do not leave an empty file under the removed name
`-y`	off	skip the start question; batches are still confirmed
`--include-kept`	off	show kept clips again
`--dry-run`	off	list candidates only

8 flatten

8.1 Concept

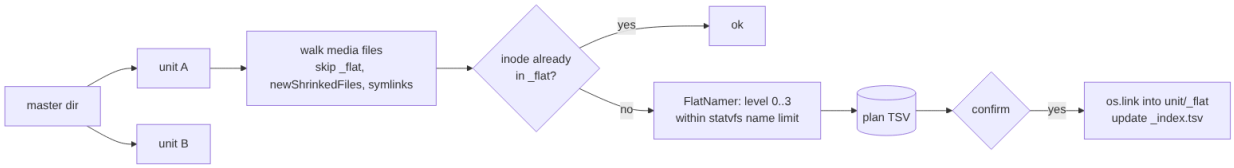


Figure 3: flatten workflow

The start directory is the **master level**; each direct sub-directory is a **unit**. Every media file below a unit (videos, images, audio; no databases) gets a hard link in `<unit>/_flat/`. Files lying directly in the unit are linked with their unchanged name, so `_flat` is complete.

```
mediatools flatten ~/Pictures/collection --dry-run
mediatools flatten ~/Pictures/collection
mediatools flatten ~/Pictures/collection --prune
mediatools flatten --undo \
  ~/scripts/mediatools.d/plans/flatten_2026-11-03_13-12-52.tsv
```

8.2 Naming

The limit is read at run time with `os.statvfs(unit).f_namemax` and counted in **bytes** (UTF-8). eCryptfs with encrypted file names reports 143 bytes on a lower file system with 255; ext4 and fscrypt report 255.

Level	Rule
0	path components joined with __, original file name unchanged
1	folder names from `[flatten] abbreviations` shortened (default `Videos` V, `Images` I, `Audios` U), `<id> <date> <time>` folders compacted to `<id>-<YYYYMMDD>T<hhmmss>`
2	level 1, then the front of the file stem is cut; the tail (usually an id) is kept; ~ marks the cut
3	directory part replaced by an 8-digit SHA-1 prefix, stem cut

Example of level 2 (limit 80 bytes, `abbreviations = posts=p, members=m, Videos=V`):

```
p__m__987654321-20240501T100000__V__~h_extended_commentary_and_extras_a1b2c3.mp4
```

In a simulation over a real collection of more than 100,000 media files with a limit of 143 bytes, more than 99.9 % of the names fitted at level 0; the rest needed level 1 or 2, none was impossible and no collision occurred. If two sources would get the same name, the second gets `~<6-digit hash>` before the extension.

The separator __ was chosen because it hardly ever occurs in real folder and file names; _ alone occurs almost everywhere.

8.3 Traceability

- `<unit>/_flat/_index.tsv` – one line `link name<TAB>source path relative to the unit`, written atomically.
- By inode, independent of any file:

```
find ~/Pictures/collection/A -samefile "$HOME/Pictures/collection/A/_flat/LINKNAME"
```

8.4 Re-runs, replaced sources, pruning, undo

Situation	Action
source inode already present in `_flat` (any name)	`ok`, nothing done
new source	`link`
same source path, new inode (file was replaced, e.g. re-downloaded)	`relink` (atomic: temporary link + rename)

Situation	Action
source gone, link still in `_flat`	`prune` – only with `--prune`
name impossible within the limit	`skip`, listed in the plan

Each run saves `plans/flatten_<ts>.tsv`. `--undo PLAN` removes exactly the links of that plan that are still identical to their source.

9 sync

9.1 Workflow

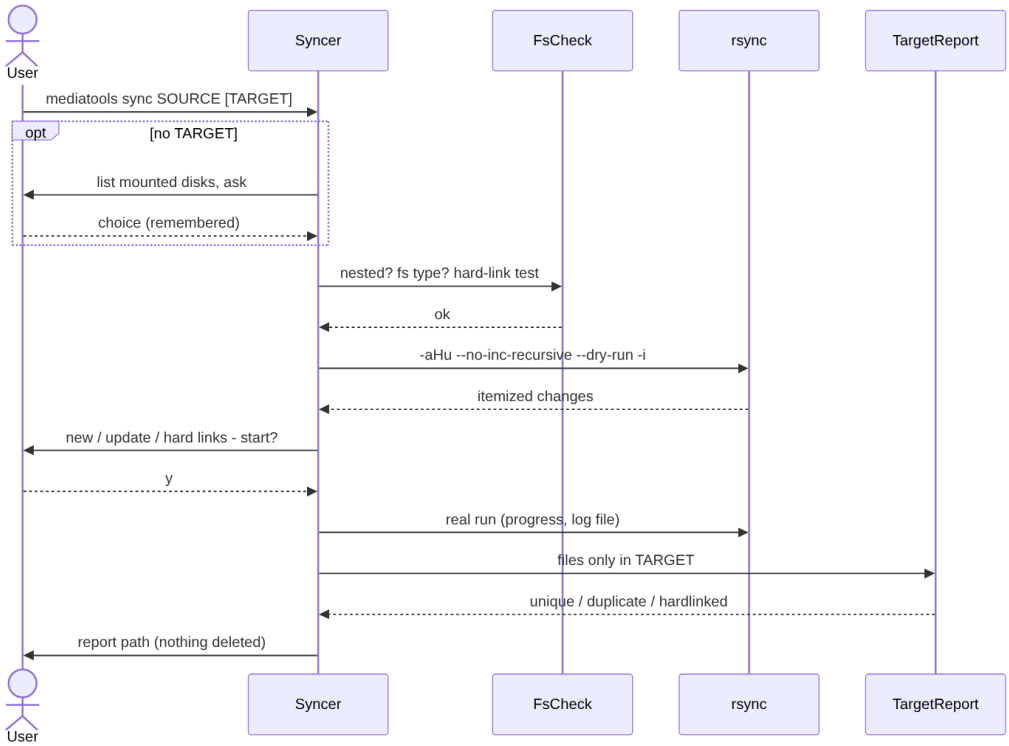


Figure 4: sync sequence

```
mediatools sync ~/Pictures/collection
mediatools sync ~/Pictures/collection /media/USER/backup/collection
mediatools sync ~/Pictures/collection --reverse
mediatools sync ~/Pictures/collection --report-only --find-duplicates
```

Without a target, mounted file systems below `/media`, `/run/media` and `/mnt` are listed with type and free space; the choice is stored in `state/sync_targets.json` and offered next time.

9.2 rsync options

```
rsync -aHu --no-inc-recursive --min-size=1 \
      --exclude=newShrinkedFiles/ --exclude=*.part \
      --exclude=.Trash-*/ --exclude=.mediatools-hl-* SOURCE/ TARGET/
```

Option	Reason
`-a`	archive mode: recursion, times, permissions, symlinks
`-H`	hard links stay hard links – `_flat` costs no space on the target
`-u`	a newer file on the receiver is never overwritten (protects shrunk files when copying back)
`--no-inc-recursive`	complete file list first; recommended by the rsync manual with `-H`
`--min-size=1`	empty triage placeholders never replace the full file on the target
no `--delete`	files that exist only in the target are kept and reported

rsync detects hard links only inside the transfer set; therefore the unit directories and their `_flat` must be in the same run, which is the case when the master directory is synchronised as a whole.

9.3 Checks before the transfer

1. rsync installed, source is a directory.
2. Source and target do not contain each other.
3. File-system type of the target; exFAT, FAT, NTFS are refused.
4. Practical test: a temporary file is created and hard-linked in the target.
5. Dry run with `--itemize-changes`; summary of new files, updates, hard links, new directories; confirmation.

9.4 Target-only report

`reports/sync_target_only_<ts>.tsv`:

Status	Meaning
`unique`	only copy anywhere – keep unless known obsolete
`duplicate`	identical content elsewhere on the target (different inode)
`hardlinked`	another name of the same inode exists on the target

Content identity: equal size, then SHA-256 of first and last MiB, then full SHA-256. With `--find-duplicates` all groups of identical files in different inodes on the target are listed – space that hard-link deduplication could reclaim.

9.5 Both directions

`sync A B` copies A to B; `sync A --reverse` (or `sync B A`) copies back. Since nothing is deleted, copying back restores files that were removed on purpose (e.g. by `triage`); the dry-run summary shows how many files would be created – decline if that is not intended.

9.6 When to use sync and when snapshot

`sync` keeps one mirror and never deletes – safe, but deletions in the source accumulate on the disk. `snapshot` (next chapter) propagates deletions to the newest snapshot while older snapshots keep the files, so a mistaken deletion stays recoverable. On a disk that is used for snapshots, `sync` should no longer write into the snapshot root.

10 snapshot

10.1 Principle

rsnapshot stores every snapshot as a complete directory tree. Unchanged files are hard links to the previous snapshot, so a snapshot costs only the changed files plus directory metadata. mediatools generates the rsnapshot configuration from the INI file (`snapshot/rsnapshot.conf` in the data directory) and adds what rsnapshot does not do by itself: checks, a dry-run probe before every change, a deletion limit, adoption of an existing copy and a schedule for a disk that is not always attached.

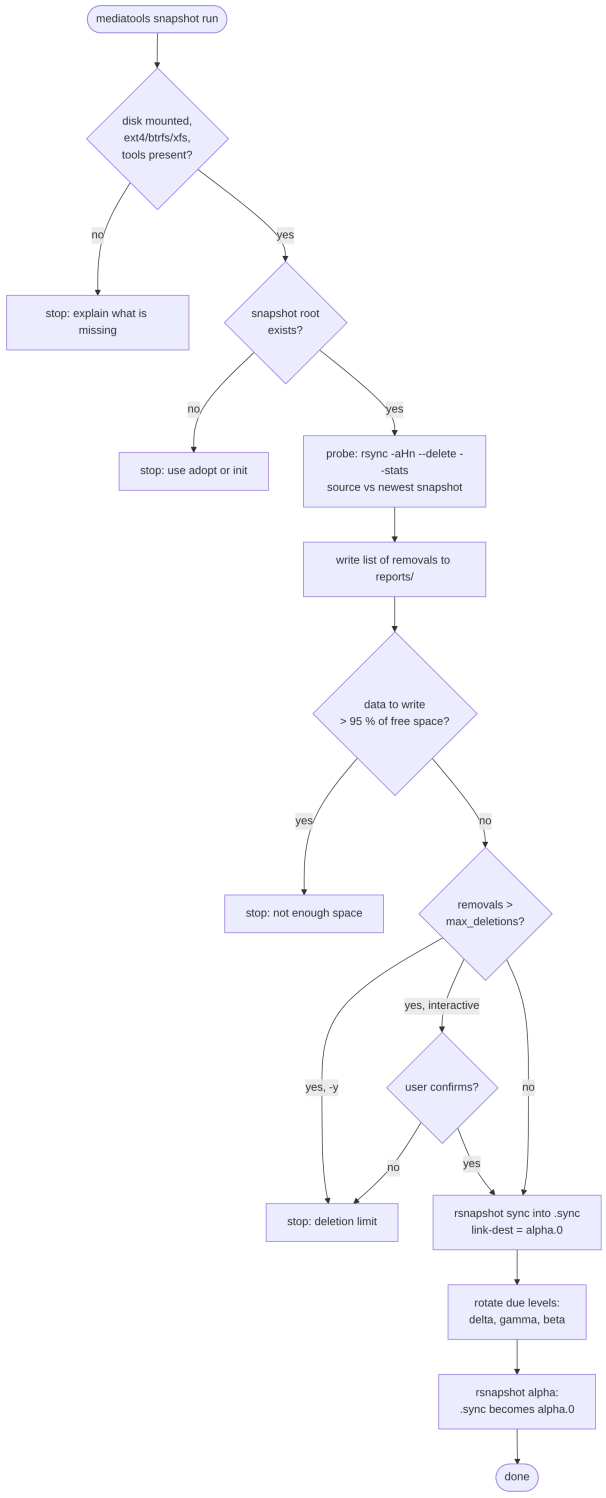


Figure 5: snapshot run

Fixed rsnapshot settings and why:

Setting	Reason
`no_create_root 1`	if the disk is not mounted, rsnapshot stops instead of filling the empty mount point on the system disk
`sync_first 1`	`rsnapshot sync` copies into `.sync`; the level commands only rotate
`link_dest 1`	unchanged files become hard links to the newest snapshot
`-aH --no-inc-recursive`	`_flat` hard links stay hard links within each snapshot
`--delete --delete-excluded`	only the newest snapshot follows deletions; older ones keep the files
`--relative`, destination `localhost/`	snapshot path is `<level>.<n>/localhost/<absolute source path>`

10.2 Commands

```
mediatools snapshot check              # read-only: config, disk, tools, probe
mediatools snapshot adopt DIR_ON_DISK  # once: existing copy = first snapshot
mediatools snapshot init               # or: empty start, first run copies all
mediatools snapshot run --dry-run      # probe only, removals listed in reports/
mediatools snapshot run                # probe, confirm, sync, rotate
mediatools snapshot run -y             # no questions; aborts instead of asking
mediatools snapshot status             # snapshots, free space, last rotations
mediatools snapshot install-trigger    # systemd user unit, started on mount
```

All commands run as normal user. rsnapshot is usually started as root by cron; with its own configuration file it can run as any user (rsnapshot manual). A normal user is sufficient as long as all files of the source belong to that user and the snapshot root on the disk is writable for him.

10.3 Taking over an existing copy (adopt)

If the disk already holds a copy of the source made with `sync`, that copy becomes the first snapshot without copying any data:

1. **Checks** – disk mounted, ext4/btrfs/XFS, directory on the same file system as the disk, snapshot root empty or missing.
2. **Probe** – `rsync -aHn --delete --stats` from the source to the given directory. If more than 25 % of the source volume would have to be transferred, the directory is not a copy of the source (wrong path or wrong level); the command stops and shows the top-level entries of both so the path can be corrected. The list of files that the first run would remove from the newest snapshot is written to `reports/`.
3. **Rename** – after confirmation the directory is renamed to `.sync/localhost/<source path>` (`rename(2)` within one file system: no data is moved, no space needed).
4. **Rotate** – `rsnapshot alpha` turns `.sync` into `alpha.0`.
5. **stand0** – `cp -al alpha.0 stand0` creates a hard-link copy that is never rotated away, so the files that the first runs remove stay available permanently. Omit with `--no-stand0`.

`stand0` must be a copy, not a renamed `alpha.0`: the next run links against `alpha.0`; without it, rsync would copy the whole source again.

10.4 Runs and levels

`snapshot run` does, in this order: checks; probe against the newest snapshot; refusal if the data to write exceed 95 % of the free space; warning and question (with `-y`: abort) if more than `max_deletions` files would leave the newest snapshot; `rsnapshot sync`; rotation of the due levels; `rsnapshot alpha`.

Level	Rotated	Kept (default)
`alpha`	every run	7
`beta`	7 days after its last rotation	4
`gamma`	30 days after its last rotation	12
`delta`	365 days after its last rotation	2

Because the disk is attached irregularly, the periods are measured from the last successful rotation (`state/snapshot_levels.json`), not from calendar dates. A higher level takes the oldest snapshot of the level below, so the first `beta` appears only once `alpha` has reached its retain count.

10.5 Starting automatically when the disk is mounted

```
mediatools snapshot install-trigger
systemctl --user daemon-reload
systemctl --user enable mediatools-snapshot.service
```

The command first asks the user's systemd instance whether the mount unit of `[snapshot] disk` is active (the disk must be mounted) and then writes `~/.config/systemd/user/mediatools-snapshot.service` with `WantedBy=<mount unit>`. The service runs `mediatools snapshot run -y` with idle I/O priority. Check afterwards:

```
systemctl --user status mediatools-snapshot.service
journalctl --user -u mediatools-snapshot.service -n 50
```

11 Testing

```
cd ~/Projects/mediatools
python3 -m pytest -q
PYTHONPATH=src:tests pylint src tests tools
```

File	Covers
`test_core.py`	media types, data directory, walker skip rules, prompt
`test_shrink.py`	database lookup, schema migration, full shrink cycle (HEVC skip, broken file, in-place inode, `_flat` skipped, DB precedence), keep-copies, duration validation, read-only progress
`test_triage.py`	kept list, player options, closed/open/early-exit classification with a stand-in player, `_flat` link resolution, removal of original plus link, no removal without confirmation
`test_flatten.py`	naming levels and byte limit, idempotency, relink, prune, undo, originals untouched

File	Covers
`test_sync.py`	hard-link test, rsync command without delete, itemize parsing, hard links preserved on the target, nothing deleted, duplicate classification, `-u` protection
`test_cli.py`	end-to-end runs of flatten, progress, sync and reverse sync
`test_config.py`	defaults, values and error messages of the INI file, commands without path, database renaming with conflicts
`test_snapshot.py`	settings validation, probe parsing, level schedule, generated rsnapshot.conf, systemd unit, refusal of a wrong directory, adoption without copying, deletion limit, removed files kept in `stand0`, hard links shared between snapshots, rotation of `beta`
`test_shrink_run.py`	run loop: two-stage Ctrl+C, SIGHUP, journal order, leftovers, forecast, HEVC set-aside, codec cache, cache checkpoints
`test_runcontrol.py`	PID and stop files, `mediatools stop` during a run and without a run
`test_results_inhibit.py`	result table layout and wrapping, report file of a run, sleep lock held/refused/missing
`test_apply_release.py`	`mediatools update` / `apply-release.sh`: newest copy per version, pending/stale, tidying of the download folder
`test_release_1213.py`	`apply-release.sh`: announcement of each CI run, summary table, failed run recorded; workflow triggers and Python matrix
`test_release_1215.py`	`mediatools status`: advice per version pair, last version line counts, pending reload/stop, missing log, source checkout, exit codes
`test_release_1214.py`	`apply-release.sh`: reading GitHub calls retried and given up after five attempts, interrupted watch resumed, red run not retried
`test_background.py`	background mode (fork, new session, console file), status file, `mediatools attach` and its Ctrl+C

The CI workflow (`.github/workflows/tests.yml`) runs all quality gates (`tools/check.sh`: format, lint, types, security, tests) on Python 3.10, 3.12, 3.13 and 3.14, for pull requests into `main` and `develop` and for release tags. Which run checks what is shown in `docs/developer.md`, section “CI runs”; the background is in `docs/ci-guide.md`. Architecture, UML diagrams and the API reference are in `docs/developer.md`.

12 Troubleshooting

Symptom	Cause	Action
`required tool(s) not found`	ffmpeg, rsync, gio or player missing	install the package named in the message
shrink: `FAILED … ffmpeg exit code`	input damaged or unsupported	`ffprobe -v error FILE`

Symptom	Cause	Action
shrink: `BROKEN, video data cannot be decoded`	sample data still encrypted (encryption boxes stripped) or scrambled; container and duration look normal	`ffmpeg -v error -t 10 -i FILE -f null -` shows the decoder errors; the file cannot be repaired without the original data
sync refuses target: `cannot store hard links`	target is exFAT/FAT/NTFS	use ext4, btrfs or XFS
sync: `not writable`	target directory owned by root	see the command below
flatten: `other file system, skipped`	file on a different mount	hard links cannot cross file systems
triage: every clip `player error`	player exits immediately	start the player manually with one file
`no path given and [collection] root is not set`	no configuration	`mediatools config --init`, set the value
snapshot: `not a mount point`	disk not attached or wrong `disk` value	`lsblk -f` shows mount points
snapshot adopt: `does not look like a copy`	wrong directory or level	compare the two top-level listings in the message
snapshot run: `aborted because of the deletion limit`	many removals (or wrong `source`)	read the list in `reports/`, then run without `-y`

```
sudo install -d -o "$USER" -g "$USER" "/media/$USER/DISK/collection"
```

`install -d -o USER -g GROUP DIR` creates the directory with the given owner; it needs `sudo` because the mount point of a disk formatted as ext4 belongs to root. Afterwards everything runs as normal user.

13 Glossary

CRF – constant rate factor; quality-based rate control of x264/x265. FFmpeg H.265 guide

eCryptfs – stacked encrypting file system; with encrypted names the limit is 143 bytes. Kernel commit “eCryptfs: Improve statfs reporting”

ffprobe – FFmpeg tool that reports stream data. ffprobe documentation

Hard link – additional directory entry for the same inode; data exist once. link(1)

HEVC / H.265 – video coding standard, about half the bit rate of H.264 at comparable quality. ITU-T H.265

Inode – file-system object holding data and metadata; names point to it.

Itemize changes – rsync output describing each change with a code such as `>f+++++++++` or `hf`. rsync manual

libx265 – open-source HEVC encoder used by ffmpeg. x265 CLI documentation

mtime – last-modification time of a file.

rsync -H – preserve hard links within the transfer set. rsync manual: –hard-links

statvfs / f_namemax – system call reporting file-system limits, including the maximum name length.

Trash (FreeDesktop) – recoverable deletion area used by `gio trash`. Trash specification 1.0

UPSERT – `INSERT … ON CONFLICT … DO UPDATE` in SQLite; needs a unique key. SQLite UPSERT

PEP 668 – marks the system Python of a distribution as "externally managed", so `pip install --user` is refused there. PEP 668

pipx – installs Python command-line applications, each into a virtual environment of its own, as a normal user. pipx

Placeholder – empty file (0 bytes) left by `triage` under the name of a discarded clip, so a download tool that skips existing names does not fetch it again.

Release tag – annotated git tag `vX.Y.Z` on `main` that marks a released state; the installer installs only such tags. git-tag(1)

rsnapshot – snapshot tool based on rsync and hard links. rsnapshot

Snapshot level – rsnapshot retention class (`alpha`, `beta`, …); each level keeps a fixed number of snapshots.

Source checkout – code run directly from a clone (launcher or `pip install -e`) instead of an installed release; `--version` names it.

systemd user unit – service managed by the user's own systemd instance, without root.

Virtual environment (venv) – self-contained Python installation directory with its own packages, created by `python3 -m venv`. venv documentation

Unit – direct sub-directory of the master directory (e.g. one project).

14 Sources

Link check of 2026-11-03. "verified" = page content retrieved on that date.

Source	Status
rsync manual source, `--hard-links` and incremental recursion text	verified
rsync manual (samba.org)	official reference, not retrieved in this session
eCryptfs statfs patch, `ENC_NAME_MAX_BLOCKLEN_8_OR_16 143`	verified
SQLite UPSERT	verified
FFmpeg H.265 guide	verified via search index (site uses bot protection)
ffprobe documentation	verified
x265 CLI documentation	verified
ITU-T H.265 (01/26)	verified
FreeDesktop Trash specification 1.0	verified
VLC command-line help	verified
link(1), man7.org	verified
`gio help trash` (libglib2.0-bin 2.80, Ubuntu 24.04)	checked locally
rsync 3.2.7 behaviour (`-H`, `-u`, itemize codes)	checked by automated tests
rsnapshot manual page (mankier.com)	found via search; `sync_first`, `link_dest`, retain levels
rsnapshot	project home page, not retrieved in this session
rsnapshot(1), FreeBSD manual pages	found via search; running as a normal user with an own configuration file
pipx documentation	found via search; `pipx install git+URL` and `pipx reinstall` checked locally with pipx 1.4.3
PEP 668	found via search

Source	Status
git-tag(1), venv documentation	standard references, not retrieved in this session; `install.sh` behaviour checked by automated tests and an end-to-end installation from a local repository
rsnapshot 1.4.5 behaviour (adopt, rotation, link sharing)	checked end to end on ext4 and by automated tests